



Robust Watermarking Against Neural Style Transfer

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Declaration of Originality

I confirm all work presented in this dissertation is the result of my independent work and investigation, except when otherwise stated.

This work has been submitted in accordance with the [following academic integrity standards](#) laid out by Liverpool Hope University.

I confirm that I understand the consequences of academic misconduct and that failure to do so may result in termination of my studies at Liverpool Hope University

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Declaration of Originality	2
Acknowledgements	2
Abstract	4
Introduction	5
Motivations	7
Related Works	7
Neural Style Transfer	7
Digital watermarking	9
Goals	11
Methodology	12
Implementation details	12
Datasets	12
Code	12
Parameters	13
Evaluation Metrics	13
Discrete Wavelet Transform (DWT)	14
Discrete Cosine Transform (DCT)	15
Watermarking	16
Adversary	18
Loss Functions	20
Perceptual Loss	20
Watermark Loss	20
Adversary Loss	20
Total Loss	21
DWT Embedding Decay Loss	21
Training Process	23
Experimentation	24
Watermarking Order	24
Watermark Resolution	24
Neural Style Transfer Model	27
Results	29
Analysis	30
Watermark	30
Styled Image	32
Future Work	35
Embedding Method	35
Improved Extraction	35
Multi-Modal Robustness	36
Adversarial Improvements	36

Invisibility Improvements	36
Robustness Against Other Distortions	36
Conclusion	36
References	37

Abstract

In the digital art era, protecting creative ownership is increasingly difficult due to widespread copying and redistribution. Traditional watermarking techniques such as Least Significant Bit (LSB), Discrete Cosine Transform (DCT), and Discrete Wavelet Transform (DWT) offer lightweight solutions for ownership verification (Cox et al. 1997; Barni et al. 1998). However, Neural Style Transfer (NST)—which applies non-linear, high-level modifications to image structure and texture—poses a significant challenge to the robustness and invisibility of these methods (Gatys, Ecker, and Bethge 2016). While robustness against compression and cropping has been widely studied, resilience under NST remains underexplored. Existing NST-resistant methods often require complex models or resource-intensive training, limiting accessibility (Zhang et al. 2024).

This project investigates whether image-based watermarks can survive NST and proposes a novel frequency-domain embedding method using hybrid DCT-DWT techniques, guided by a Generative Adversarial Network (GAN). The Adaptive Instance Normalization (AdaIN) model (Huang and Belongie 2017) is used to perform NST, acting as an adversarial transformation during training. The system is trained end-to-end, with a classifier adversary predicting watermark presence post-NST, encouraging semantic retention. A custom DWT Embedding Decay Loss prioritises embedding into low-frequency bands, aligning with NST’s disruption of high-frequency features (Simonyan and Zisserman 2015). Evaluation metrics include PSNR, SSIM, perceptual difference, and extraction accuracy.

Results show substantial watermark degradation post-NST, despite relatively high PSNR (52.68 dB), with low SSIM (0.28) indicating weak structural retention. Residual difference images reveal significant distortion and watermark “leakage” into stylised content, illustrating the trade-off between invisibility and robustness. Compared to prior work such as Zhang et al. (2024), metric divergence suggests that NST watermark evaluation depends heavily on the balance between structural fidelity and perceptual subtlety.

In summary, this project contributes an accessible, partially NST-resilient watermarking method, providing insights into the fragility of traditional techniques under non-linear style transformations, and laying groundwork for future advances in content protection.

Introduction

In the evolving digital art landscape, creative ownership has become increasingly difficult to validate online. For many independent creators and illustrators who share their work publicly on online platforms such as X (Formerly Twitter), Instagram, Artstation, and Tumblr, proving ownership is crucial, as online media is prone to illegal copying, transformation, and redistribution without accreditation.

This is often solved through traditional watermarking schemes such as LSB, DCT (Barni et al., 1998), and others, providing simple, low-cost methods to ensure owner attribution; however, with the recent introduction of machine learning technologies such as Neural Style Transfer (NST), allows for the transference of an image's style onto another (Gatys, Echer, and Bethge, 2016), which threatens both the robustness and imperceptibility of existing watermarking techniques.

Whilst significant progress has been devoted to developing techniques that withstand conventional distortions such as compression, resizing, and cropping, far less literature exists concerning watermark robustness against NST (Zhang et al., 2024; Li et al., 2024). Unlike traditional distortions, Neural Style Transfer does not apply linear degradation or distortions; rather, it introduces substantial and non-linear modifications across an image's spatial and frequency domains, significantly complicating watermark preservation, as demonstrated in **Figure 1**, reproduced from Figure 2 of Luan et al. (2017), whose main contribution was to constrain the transformation from input to output to be locally affine in colorspace, successfully suppressing distortion.

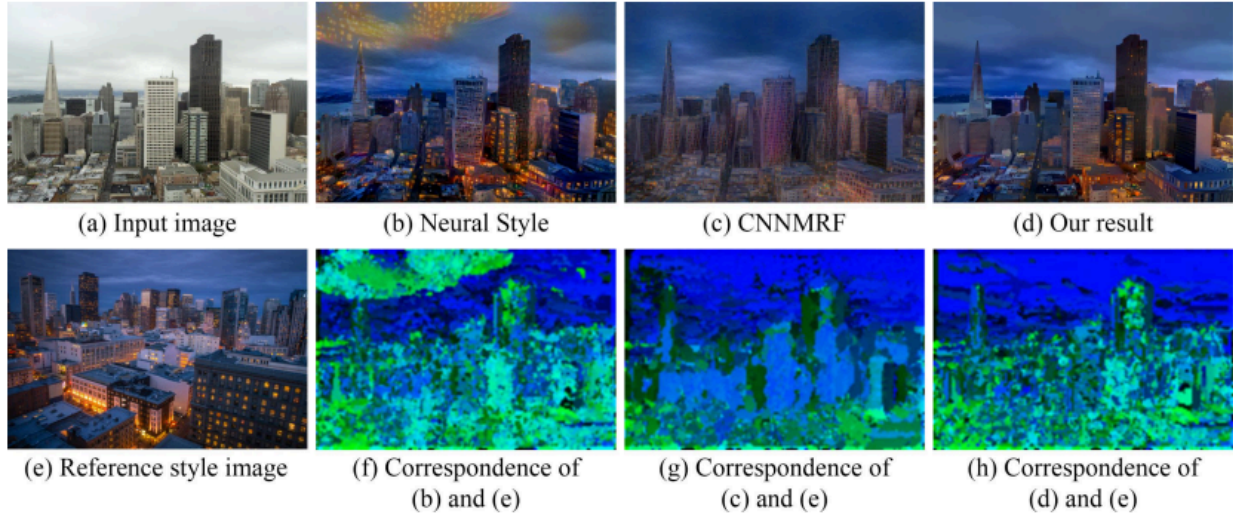


Figure 2: Given an input image (a) and a reference style image (e), the results (b) of Gatys et al. [5] (Neural Style) and (c) of Li et al. [10] (CNNMRF) present artifacts due to strong distortions when compared to (d) our result. In (f,g,h), we compute the correspondence between the output and the reference style where, for each pixel, we encode the XY coordinates of the nearest patch in the reference style image as a color with $(R, G, B) = (0, 255 \times Y/\text{height}, 255 \times X/\text{width})$. The nearest neural patch is found using the L2 norm on the neural responses of the VGG-19 *conv3_1* layer, similarly to CNNMRF. Neural Style computes global statistics of the reference style image which tends to produce texture mismatches as shown in the correspondence (f), e.g., parts of the sky in the output image map to the buildings from the reference style image. CNNMRF computes a nearest-neighbor search of the reference style image which tends to have many-to-one mappings as shown in the correspondence (g), e.g., see the buildings. In comparison, our result (d) prevents distortions and matches the texture correctly as shown in the correspondence (h).

Figure 1. Example of distortions caused by Neural Style Transfer. Reproduced from Luan et al. (2017, Figure 2)

These existing approaches additionally often involve complex architectures or computationally expensive training pipelines that are difficult to replicate on a small scale. For instance, Stylemark (Zhang et al., 2024) requires a multi-stage training process totalling over 4,000 iterations using a high-end NVIDIA RTX 3090 GPU, and 2,000 style-content image pairs. Such setups are often unavailable to the average independent developer or academic. As a result, these approaches have seen limited adoption within the wider research and creative communities, likely due to their computational demands and the absence of accessible implementations.

Given this growing threat of NST, and its potential for malicious watermark removal, there is a clear need to develop more robust watermarking techniques that are resilient to these stylistic transformations, whilst remaining accessible to other researchers.

This study aims to address this gap by proposing a novel method that embeds a watermark using a hybrid Discrete Wavelet Transform - Discrete Cosine Transform (DWT - DCT) approach, aided by a Generative Adversarial Network (GAN). The watermark embedding is controlled via adjustable DCT and DWT alpha parameters, which controls the watermarks embedding strength. Additionally, to enhance robustness against NST, the watermark is passed through the ADAIN network (Huang and Belongie, 2017), serving as an adversarial process, distorting the

image's style and watermark, to optimise against. By treating NST as an adversarial attack against watermark visibility, this research hopes to offer insights into how existing and new watermarking methods withstand style-driven manipulations.

Motivations

As an artist who frequently shares work on public and semi-public platforms, such as X and Discord, I am acutely aware of the challenges plaguing creator ownership in these communities. Platforms like these are often casual, promoting rapid sharing and discussion of content within public channels; and despite the widespread issue of art theft, watermarking is often dismissed as being overly cautious or unnecessary. This cultural attitude contributes to an environment where plagiarism, especially of sketches, works in progress, and stylistic design, can continue to flourish.

At the same time, machine-learning tools, like Neural Style Transfer are becoming increasingly popular. Despite their potential for accidental and malicious watermark removal, these tools offer genuine creative benefits, letting artists quickly iterate on visual style, palette, and mood, which makes them a powerful tool for artists.

This disconnect between the prevalence of plagiarism, the emergence of NST, and the perceived effort of watermarking highlights a real need for tools focused on the artist, notably it should offer a lightweight, unobtrusive protection strategy, and should be easy to use and invisible to audiences.

While research does exist on this topic (Zhang et al., 2024; Li et al., 2024), their implementations are computationally complex and have no code available, leaving a significant gap in the protection available to an artist. The development of a lightweight, robust, NST-resilient watermark would benefit both academia and real-world artists.

Related Works

Neural Style Transfer

In the seminal work by Gatys, Ecker and Bethge, (2016) Neural Style Transfer (NST) was introduced as a method for synthesising visually compelling images by separating and recombining the content and style representations of arbitrary inputs. Their approach employed a pretrained VGG-19 convolutional neural network to extract hierarchical feature representations of the image. Content features could then be extracted from deeper layers of the network, while style was modelled using the Gram matrix of feature activations from multiple layers, capturing the spatial correlations and texture statistics of the style image. The style transfer process involved optimising a randomly initialised (noise) image to minimise a weighted sum of two losses: content loss, computed as the mean squared error (MSE) between the noise

image and the content image's feature maps, and style loss, calculated as the MSE between the Gram matrices of the noise image and the style image (Gatys, Ecker and Bethge, 2016). While this technique produced high-quality results, its dependence on iterative backpropagation made it computationally expensive, and thus unsuitable for real-time applications.

To help address this limitation, Johnson, Alahi, and Fei-Fei (2016) introduced a new feed-forward based network, built to approximate the results produced by Gatys et al optimisation approach. This shift in architecture provided the researchers a speed 3 orders of magnitude faster than Gatys et al approach, however came at the cost of training a separate network for each distinct style.

Huang and Belongie (2017) later introduced Adaptive Instance Normalisation (AdaIN), a further departure from the optimisation-based framework of Gatys et al. Unlike previous approaches that required a dedicated model per style or a fixed set of affine parameters for normalisation layers, AdaIN enabled arbitrary style transfer in real-time by dynamically aligning the mean and variance of content feature activations to those of the style image. Specifically, AdaIN adjusts the statistics of the content features such that the resulting feature maps match the style image in distribution, effectively producing a target representation that fuses content structure with stylistic patterns. This process eliminates the need for predefined affine parameters in the instance normalisation layers, allowing a single network to generalise across unseen styles at inference time, eliminating the need for individual style training as seen in previous IN and BN approaches (Ulyanov et al., 2016). Additionally, the authors demonstrated that Instance Normalisation outperformed Batch Normalisation for style transfer tasks, resulting in lower style loss, likely due to its ability to preserve better instance-specific features critical to stylisation (Huang and Belongie, 2017).

Some later advancements include Contrastive Arbitrary Style Transfer (CAST) introduced by Zhang et al. (2022). The proposed work aims to overcome the limitations of earlier NST models that rely on second-order statistics such as Gram matrices (Gatys, Ecker and Bethge, 2016) and channel-wise variance (Huang and Belongie, 2017), which is addressed by learning style representations directly from image features using a contrastive learning paradigm. A key component of the paper is the Multi-layer Style Projector (MSP), which extracts features from a fine-tuned VGG19 model to encode both global and local style characteristics into a style code, which can then be used for other models such as ADAIN, replacing the mean and variance step used.

To enable this, CAST employs a contrastive learning framework where an image and its augmented counterpart (created via random cropping, resizing, and rotation) are both passed through the style encoder to produce feature embeddings. These embeddings are then projected into a compact vector space, where similarity is maximized between the augmented pair (positive example) while being minimized against unrelated images (negative samples) drawn from a large memory bank. This contrastive loss encourages the network to learn discriminative and semantically meaningful style codes, enhancing its ability to generalise across diverse artistic domains and preserve stylistic identity more robustly than previous methods.

To address uneven and distorted styling that can be seen in models such as ADAIN, it also implements a Domain Enhancement Scheme that utilises an adversarial loss designed to minimise content loss, ensuring a more even styling application. These components collectively allow the model to learn and manipulate style distributions better than its competitors. Empirical results, including LPIPS scores, content preservation metrics, and user studies, demonstrate CAST's better outputs, superior to former methods in both visual quality and style transfer capabilities.

Digital watermarking

Digital watermarking has been a foundational component to image security for over thirty years, with early contributions (Cox et al., 1997; Swanson et al, 1998; Barni et al., 1998) establishing the three main objectives a watermark should strive for: imperceptibility, robustness, and embedding capacity.

Initial methods focused on directly modifying pixel values to encode information; however, this was shown to be highly fragile against common distortions such as compression or resizing.

Frequency-domain techniques such as Discrete Cosine Transform (DCT) and Discrete Wavelet Transform (DWT) were developed to address the limitations of early spatial-domain watermarking. By embedding the watermark into the frequency domain, the watermark achieved improved robustness and imperceptibility (Swanson et al., 1998)

DCT-based methods (Barni et al., 1998) exploit the fact that human perception is more sensitive to low-frequency distortion, as that is where most visual information is stored. This leaves a large embedding capacity within mid- and high-frequency regions. By embedding in mid-frequency regions, the watermark remains imperceptible to human vision yet is more robust against compression and other distortions.

Similarly, DWT-based techniques would decompose the content into multiple frequency sub-bands (Xia, Boncelet, and Arce, 1998), allowing for watermarking across several bands (LL, LH, HL, HH) and providing greater flexibility in the embedding strategy used.

Whilst typical frequency-domain techniques were capable for their time, offering strong robustness against typical distortions, the emergence of machine learning has introduced a new era of watermarking strategies. In particular, learning-based variants of DCT and DWT have been proposed to improve watermark robustness, invisibility, and adaptability.

For example, Kavitha, Eranna, and Giriprasad (2020) proposed a method that uses a strong DCT-DWT hybrid algorithm to embed an invisible and visible watermark into a content image, then deploying a neural network to extract the watermark without access to the original pure image. Kulkarni and Kuri (2017) also present a similar blind digital image watermarking technique, utilising a DCT-DWT algorithm and a Probabilistic Neural Network (Specht 1990) to achieve better imperceptibility and robustness. These approaches represent a fundamental shift from fixed-rule embedding schemes to dynamic, task-specific models.

However, this same area of research has exposed new vulnerabilities as demonstrated by Zhao et al. (2023), who show that generative models - particularly diffusion based models - can be

used to launch regeneration attacks. These attacks work by injecting noise into an image, then reconstructing a clean version to effectively remove an embedded watermark. Whilst not directly focused on Neural Style Transfer, their findings highlight how a deep learning model can be used to adversarially distort an images underlying features to remove a watermark. In the case of NST, the process is shown to alter both spatial and frequency domains during the transference of stye from one image to another, which, much like the attack shown in Zhao et al study, can disrupt and fully remove frequency based watermarks including using SVD, DCT, DWT (Navas et al., 2008).

In response, recent research has begun to explore watermarking approaches that remain robust against Neural Style Transfer. Zhang et al (2024) proposed Stylemark, a new style watermark network that embeds the watermark into the multi-scale shared style feature space from the style image. The core concept behind Stylemark is taken from the observation that during NST, stylistic features such as brushstrokes, are extracted from the style image, whilst structural and semantic information is taken from the content image, hypothesising that watermark information could be embedded into the shared style features of the style image, which can then be co-rendered to the final NST processed image.

To achieve this the model extracts multi-scale features from the style image and incorporates the watermark into these features layer by layer, additionally it implements a distribution squeeze loss, an additional constraint designed to minimise statistical feature distortion, this loss effectively forces the encoder to embed most watermark information into redundant style features, which are more likely to be preserved and transferred by NST. Their experimental results demonstrate significant robustness against several black box Arbitrary Style Transfer (AST) implementations, achieving an average watermark accuracy of 0.942, as compared to the 2nd best, MBRS (Jia, Fang and Zhang, 2021) with 0.776.

A related, but distinct approach can be seen in HIDDEN (Zhu et al., 2018), which introduces an end-to-end trainable framework built for data hiding. By leveraging neural networks' capabilities to learn and embed imperceptible perturbations, the network is capable of embedding meaningful information into the image without compromising visual fidelity. To enhance robustness for watermarking, a noise layer is optionally inserted that can mimic real-world distortions such as Dropout, cropping, Gaussian blur, or JPEG compression, along with the ability to insert additional stages to its training pipeline. This is used adversarially and fed back into the encoder. In principle, allowing an NST transformation to be inserted into its training pipeline to enhance robustness against NST.

Results show that HiDDeN performs competitively against existing watermark approaches, often surpassing commercial baselines like Digimarc under spatial noise.

Complementing this line of research, is a different line of content protection as proposed by Li et al (2024), who introduce Neural Style Protection (NSP). Unlike the approaches seen by Stylemark and Hidden, which aim to covertly embed a watermark, NSP approaches the problem adversarially, attempting to actively sabotage the NST process to produce a degraded final image, deterring unauthorised use. This is achieved by introducing imperceptible perturbations to the images intermediate feature representations, significantly changing how an NST model

“sees” the image, resulting in an output that is noticeably more distorted than baseline. To maximise the models effectiveness, NSP additionally utilises a momentum-based ensemble method that aggregates gradients from the encoders of several NST models, including ADAIN (Huang and Belongie, 2017) and CAST (Zhang et al., 2022), to allow for a more general solution that remains effective across both known and unknown NST models.

NSP, while effective in its domain, focuses on disrupting style transfer rather than preserving visual integrity. Its adversarial approach, though valuable for content protection, contrasts with this project’s goal of maintaining image fidelity and creator intent. Thus, NSP’s destructive methodology is less suited to artist-focused use cases that prioritize unobtrusive and reversible protection.

This trend towards learning-based watermarking motivates the approach taken in this project, which aims to leverage the strengths of traditional frequency-domain techniques within a deep learning framework tailored for robustness against Neural Style Transfer. By combining the proven resilience of transforms such as DWT and DCT with the adaptability of neural networks, the goal is to develop a hybrid model capable of embedding watermarks that survive stylistic transformation while preserving image fidelity. Building on the insights of works like HiDDeN and Stylemark, this project positions NST as both a threat model and a benchmark, using it to evaluate the watermark’s visibility, recoverability, and resistance under challenging real-world scenarios.

Goals

The primary goal of this project was to investigate whether digital watermarks can survive Neural Style Transfer (NST) transformations, and to develop a watermarking system that balances robustness and invisibility under these conditions. Unlike traditional watermarking threats such as cropping or compression, NST poses a unique challenge by fundamentally altering an image’s structure and frequency content through non-linear feature recombination.

To address this, the project set out the following key objectives:

- Design a hybrid watermarking method that embeds image-based watermarks into the frequency domain using a combination of Discrete Wavelet Transform (DWT) and Discrete Cosine Transform (DCT).
- Evaluate watermark resilience against NST using both pixel-level extraction accuracy and a trained adversarial detection network that treats watermark presence as a binary classification task.
- Quantitatively assess invisibility using metrics such as PSNR, SSIM, and perceptual difference, to ensure that the watermark does not degrade visual quality.

Methodology

Implementation details

Datasets

Content images were obtained from the flickr8k (adityajn105, 2020) dataset, as the dataset features a multitude of natural images depicting a variety of scenes and scenarios, which provide a diverse range of content images to test on.

For styling images, a separate dataset, “Best Artworks Of All Time” (Icaro, 2019), was chosen due to the striking visuals provided by the 50 most influential artists of all time, as described by the author.

Alongside this, the “Lena” test image is also utilised for comparison means against other papers and implementations.¹

Due to hardware limitations, a random selection of 50 image pairs were taken from each dataset respectively for training, additionally, all images were resized to 224 x 224 to match the VGG19 (Simonyan and Zisserman, 2015, sec. 2.1) expected input size.

The watermark, **Figure 2** is used throughout the project is a 16x16 mono-colour image, resized to 224 x 224. Providing a maximum hypothetical 30 bits of embedding information capable of representing just over 1 billion possible watermarks. However, to ensure consistency and comparability across experiments, a single static watermark was used throughout the study.

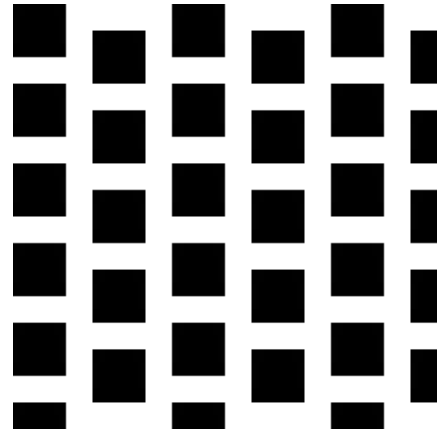


Figure 2: Watermark Image

Code

The implementation was performed using Pytorch and executed locally on an NVIDIA GTX 1060, 6GB model.

Additionally, the libraries torch_dct (zh217, 2018) and pytorch_wavelets (fbcotter, 2019) were used to implement hardware-accelerated Discrete Cosine Transform and Discrete Wavelet Transform, respectively. These libraries offer native tensor-based implementations that integrate seamlessly with PyTorch's autograd system, enabling in-place GPU computation without

¹ The Lena image is widely used in image processing research, originating from a 1972 issue of *Playboy*. Despite ongoing ethical discussions about its use, it remains a de facto standard for benchmarking due to its texture and detail diversity. As such the image is used throughout this paper.

breaking the computational graph, thereby preserving gradient flow and ensuring an uninterrupted training process.

Earlier attempts using NumPy-based approaches were shown to significantly degrade performance, primarily due to the lack of hardware acceleration and the need to convert between Pytorch tensors and Numpy arrays.

Parameters

Iterations: A fixed iteration count of 25 was used, whilst a higher value would be preferred, training time was limited due to time constraints and computational resources.

Learning rate: A learning rate of 0.001 was empirically set for both the generator and detector, based on preliminary research.

The following hyperparameters were used throughout training to control various aspects of the watermark embedding and style transfer process.

Pixel Loss Weighting: A weight of 0.3 was applied to the pixel loss term, which measures the difference between the watermarked image and its stylised counterpart. This weight helps balance the preservation of visual quality while incorporating the watermark.

Adversarial Loss Weighting: The adversarial loss, which assists in detecting the watermark through a discriminator network, was given a weight of 0.5. This ensures that the watermark detection is effective without overly distorting the content of the image.

Watermark Loss Weighting: The mean squared error (MSE) loss between the original watermark and its extracted version was also weighted at 0.5. This weight encourages accurate extraction of the watermark while maintaining a high level of quality in the watermarked image.

DWT Weights: A weight of 0.8 was applied to the Discrete Wavelet Transform (DWT) frequencies to preferentially decrease the impact of higher-frequency sub-bands during the embedding process. This weight follows a geometric decay (e.g., 1.0 for the lowest frequency band, 0.8 for the next, 0.64 for the one after, etc.), ensuring that the watermark is embedded primarily in lower-frequency regions that are less perceptible to human eyes.

Evaluation Metrics

To ensure proper analysis and evaluation of results, a mixture of quantitative and qualitative metrics were used, notably PSNR, SSIM (Wang et al., 2004) to compare the quality of the watermark produced.

In addition to these quantitative metrics, perceptual loss was included in the evaluation, as

shown in **Equation 1**. The perceptual loss is designed to capture the high-level feature differences between the input images by comparing their feature map activation. Unlike pixel-wise loss functions, which may not always align with human visual perception, perceptual loss uses feature representations from a pre-trained network, VGG-19 (Simonyan and Zisserman, 2015) in this study, to assess the similarity of image structures in a way that more closely reflects perceptual differences. This makes it particularly useful for evaluating the perceptual quality of the watermark and its potential distortion.

Discrete Wavelet Transform (DWT)

This project integrates the Discrete Wavelet Transform (DWT) as a core component of the watermarking pipeline. During DWT decomposition, the image is decomposed into a hierarchical signal representation through progressive filtering and downsampling, producing several frequency bands: LL, LH, HL, and HH, which encode different levels of frequency and directional information.

The most common DWT algorithm used is the Haar algorithm, which applies a low-pass (scaling function) and a high-pass (wavelet function) along the rows and columns of an image to produce 2 bands containing the approximation and detail coefficients, respectively.

Mathematically, this can be expressed as:

$$LL[k] = \frac{x[2k] + x[2k + 1]}{2} \qquad HH[k] = \frac{x[2k] - x[2k + 1]}{2}$$

This process can then be repeated along the other dimension of the image to produce the four distinct bands wherein The LL band captures the coarsest spatial granularity, encoding low-frequency content that often reflects the image's general structure, and significantly contributes to the final image representation, whilst the higher frequency bands contain edge and texture information, holding less information.

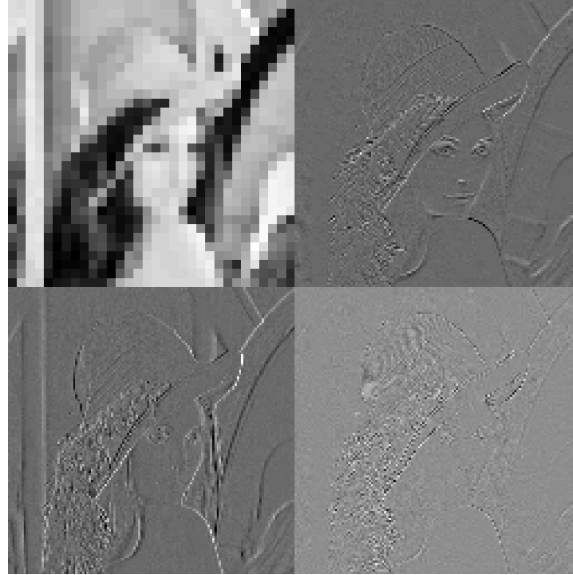


Figure 3: LL, LH, HL, HH DWT sub-bands (left - right, top-bottom)

Interestingly, this separation of spatial information from DWT closely matches the layout of feature representations taken from VGG19. For instance, embedding within the LL band, which captures a majority of the structural information, closely matches more activation in lower convolutional layers (e.g., conv3_1, conv3_2), which is likely due to VGG19's lower feature layers, capturing more structural information as well, which is supported by observations from Gatys et al. (2016) seminal work.

This is important for when aiming for robustness against Neural Style Transfer models such as AdaIN (Huang and Belongie, 2017) and Gatys et al (2016), along with other models that utilise VGG19 for feature extraction and encoding, as they often only extract from lower layers (up to relu4_1).

As a result, embedding watermark information into lower DWT bands, specifically LL, increases the likelihood that this information is retained post Neural Style Transfer. This coincidental alignment between frequency-domain embedding and feature maps forms a key principle for the design of this project.

Discrete Cosine Transform (DCT)

The Discrete Cosine Transform is also used in the implementation alongside DWT to produce a hybrid algorithm. The technique allows for expressing an image as a sum of cosine functions oscillating at different frequencies.

In the context of image processing, the 2D DCT (Barni et al. 1998) is applied to blocks of pixels, commonly 8x8 or 16x16, transforming the pixels' spatial intensity into frequency coefficients. The resulting block is composed of DC coefficient at the top left, representing the average colour or intensity of the image, followed by AC coefficients that encode the spatial variance of

the image, often containing edge and fine texture detail. Mathematically, the 2D DCT can be expressed as:

$$F(u, v) = \left(\frac{2}{N}\right)^{\frac{1}{2}} \left(\frac{2}{M}\right)^{\frac{1}{2}} \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} \Lambda(i) \Lambda(j) \cdot \cos\left[\frac{\pi u}{2N} (2i + 1)\right] \cos\left[\frac{\pi v}{2M} (2j + 1)\right] \cdot f(i, j)$$

(Marshall, 2001)

Where $\alpha(u)$, $\alpha(v)$ are normalisation factors based on u, v

Because the DCT concentrates most of the signal energy into a few low-frequency components, it enables effective compression and resilience. In this watermarking context, embedding is performed in mid-frequency coefficients, which balances robustness (surviving compression or NST) and imperceptibility (avoiding visible distortions). This choice ensures that watermark data remains hidden while being resistant to common image transformations.

Watermarking

The core watermarking process used is a hybrid DCT-DWT embedding pipeline designed for robustness against NST attacks. This section outlines the pipeline's DCT-DWT watermarking component.

Before embedding, the watermark is resized to match the image's shape and channel count if needed.

For DWT embedding, the `pytorch_wavelets` library is used to allow for tensors.

Although the LL band is the main target for embedding due to its structural importance and alignment with low-level VGG19 features, the algorithm also embeds watermark data into the higher-frequency bands. This is done using smaller alpha scaling factors to reduce visible distortion. As a result, all bands contribute to the embedded watermark, but the LL band carries the most significant portion.

After applying the DWT, the inverse DWT is applied to reconstruct the watermarked image.

Algorithm 1: DWT Embedding

Input:

- Content Image (content)
 - Watermark Image (watermark)
 - DWT alphas (alphas)
-

Output:

- DWT Watermarked Image
-

```

IF SHAPE(content)  $\neq$  SHAPE(watermark) THEN
    watermark    Resize watermark to content SHAPE
END IF
IF CHANNELS(content)  $\neq$  CHANNELS(watermark) THEN
    watermark    Repeat watermark across CHANNELS
END IF
(YL_con, YH_con)  DWT(content)
(YL_wm, YH_wm)   DWT(watermark)

IF SHAPE(YL_con)  $\neq$  SHAPE(YL_wm) THEN
    YL_wm    Resize YL_wm to YL_con SHAPE
END IF
new_YH_wm    []

FOR i FROM 0 TO LENGTH(YH_con) - 1 DO
    con_coeff  YH_con[i]
    wm_coeff   YH_wm[i]

    IF SHAPE(con_coeff)  $\neq$  SHAPE(wm_coeff) THEN
        wm_coeff  Resize wm_coeff to con_coeff
                SHAPE
    END IF

    APPEND wm_coeff TO new_YH_wm
END FOR

YL_emb  YL_con + dwt_alphas[0]  $\times$  YL_wm

YH_emb  []
FOR i FROM 0 TO LENGTH(YH_img) - 1 DO
    a_idx  MIN(i + 1, 3)
    emb    YH_img[i] + alphas[a_idx]  $\times$  new_YH_wm[i]
    APPEND emb TO YH_emb
END FOR

watermarked  IDWT(YL_emb, YH_emb)

watermarked  CLAMP(watermarked, 0, 1)

RETURN watermarked

```

Following this, the watermark is also embedded within the DCT coefficients.

Algorithm 2: DCT Embedding

Input:

- Content Image (content)
 - Watermark Image (watermark)
 - DCT alpha (alpha)
-

Output:

- DCT Watermarked Image
-

```
IF SHAPE(image)  $\neq$  SHAPE(watermark) THEN  
    watermark    Resize watermark to image SHAPE  
END IF
```

```
IF CHANNELS(image)  $\neq$  CHANNELS(watermark) THEN  
    watermark    Repeat watermark across CHANNELS  
END IF
```

```
image_dct    DCT(image)  
watermark_dct    DCT(watermark)
```

```
watermarked_dct    image_dct + alpha  $\times$  watermark_dct
```

```
watermarked    IDCT(watermarked_dct)
```

```
watermarked    CLAMP(watermarked, 0, 1)
```

```
RETURN watermarked
```

The final step applies the inverse DCT (IDCT) to reconstruct the the final watermarked image.

Adversary

To ensure robustness against NST, an adversarial detection network is used to assess whether a watermark is still present in the image post NST. Unlike traditional extraction pipelines that attempt to reconstruct the embedded watermark to assess robustness, this module treats detection as a binary classification task: Given the original, none watermarked image, and the final stylised, watermarked image, the network attempts to predict whether the embedded watermark signal is still present.

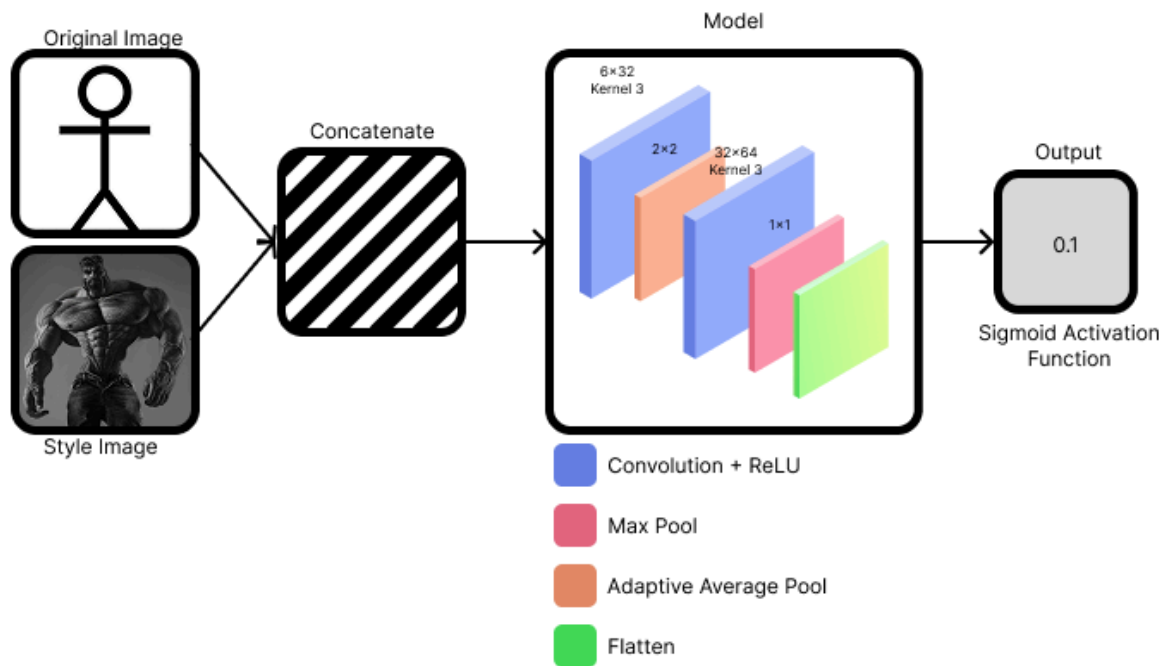


Figure 4: Adversary detection architecture

The architecture consists of a lightweight convolutional classifier containing the following layers:

- **Conv2d(6 → 32)** A convolution layer processes the 6-channel input, extracting low-level differences between the original and watermarked image.
- **ReLU + MaxPool2d(2)** Applies non-linearity and downsampling, helping the model focus on localized changes.
- **Conv2d(32 → 64)** Another convolutional layer increases representational depth, allowing the model to learn more complex patterns.
- **AdaptiveAvgPool2d((1,1))** Globally averages the feature map to a single spatial value per channel, reducing it to a 64-element vector.
- **Flatten + Linear(64 → 1)** Fully connected layer compresses the features into a single scalar.
- **Sigmoid** Outputs a probability between 0 and 1 indicating whether a watermark is detected.

Integrating the adversary into the pipeline provides a practical signal of real-world robustness — rather than simply measuring pixel-wise recovery, it evaluates whether a downstream system can reliably detect the watermark under stylistic distortion. This feedback can also be used adversarially during training, guiding the encoder to produce more resilient embeddings.

Loss Functions

Perceptual Loss

The perceptual loss is used to assess the visual loss of the styled image. By comparing the feature activations, obtained from a VGG19 feature extractor, between the base-styled image, and the styled, watermark image, this loss calculates how visible the watermark is in the final image.

Perceptual loss is used as it is shown to better represent how people see an image, and can result in a better output when compared to pixel loss. (Grund, Sandin, and Liwicki 2020)

$$Loss_{perceptual} = \frac{1}{N} \left\| \hat{F}_1 - \hat{F}_2 \right\|$$
$$\hat{F}_x = \text{concat}(\text{flatten}(F_x^1), \text{flatten}(F_x^2), \dots, \text{flatten}(F_x^{end}))$$

Equation 1: Perceptual Loss

Watermark Loss

Watermark loss quantifies the pixel-level difference between the original and extracted watermarks after the NST process. It is computed as the mean absolute error (MAE) across all pixels, penalizing distortion or loss of watermark fidelity during transformation.

$$Loss_{watermark} = \frac{1}{N} \sum_{i=1}^N \left| watermark_{original} - watermark_{extracted} \right|$$

Equation 2: Watermark Loss

Typically, pixel loss is used to help preserve the visual similarity between images; in this context, the same loss serves a different purpose. Since the watermark utilises a hybrid DCT DWT embedding algorithm, the pixel loss between the watermark is indirectly measuring how much frequency information survives the NST transform. Proxying NSTs frequency-space information retention, ensuring the watermark stays “visible” within the image’s frequency domain even after stylisation. This differs from the approach taken by Stylemark (Zhang et al., 2024), which utilises Binary Cross-Entropy (BCE) to compare binary watermark predictions. Whilst BCE is effective when using a binary sequence watermark, this project utilises an image for the watermark, making pixel loss more suitable for classifying watermark robustness.

Adversary Loss

An adversarial loss is introduced to complement pixel-level tracking, computed via binary cross-entropy from a lightweight convolutional discriminator. Rather than reconstructing the watermark, this adversarial module classifies whether a watermark is still present in the stylised image.

This loss introduces a robustness signal: if the adversary can no longer detect the watermark after stylisation, it signals failure. The detection mechanism is explained in more detail in the separate adversarial architecture section. Combined with pixel loss, the adversarial signal encourages the system to embed watermarks that are both imperceptible and resilient to neural style transfer.

$$Loss_{adversary} = -\frac{1}{N} \sum_{i=1}^N \log D(x_i)$$

Where $D(x_i)$ is the adversaries prediction

Equation 3: Adversary Loss

Total Loss

The final loss is a weighted sum of the three components:

- **Perceptual loss** encourages visual fidelity
- **Watermark loss** encourages reconstructability in the frequency domain
- **Adversary loss** enforces semantic detectability despite transformations

Together, they guide the embedding network to balance visibility, robustness, and imperceptibility — qualities often at odds in watermarking tasks.

$$Loss_{Total} = (\lambda_{perceptual} \cdot Loss_{perceptual}) + (\lambda_{watermark} \cdot Loss_{watermark}) + (\lambda_{adversary} \cdot Loss_{adversary})$$

Equation 4: Total Loss

DWT Embedding Decay Loss

In most traditional watermarking studies (Xia, Boncelet, and Arce, 1998), the LL band of DWT is often avoided due to its greater perceptual impact, often preferring middle frequencies as they provide a good balance between visibility and robustness. This project takes the opposite approach, intentionally emphasising the LL band.

This is because in the context of Neural Style Transfer, the watermarks robustness depends heavily on where in the feature space the watermark is embedded, through experimentation it was shown that the LL band promotes greater embedding within the first few feature activations of a VGG19 model, an area which NST distorts less, whereas the higher layers are often

discarded or greatly distorted as this is where style information is stored. Therefore, a watermark embedded within lower-frequency components is more likely to survive NST.

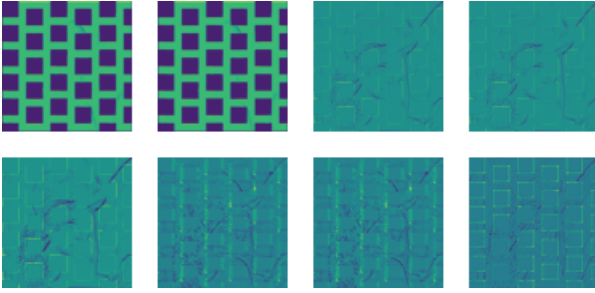


Figure 5: First 8 layers of VGG19 activation layers difference.
LL band - 1.0.

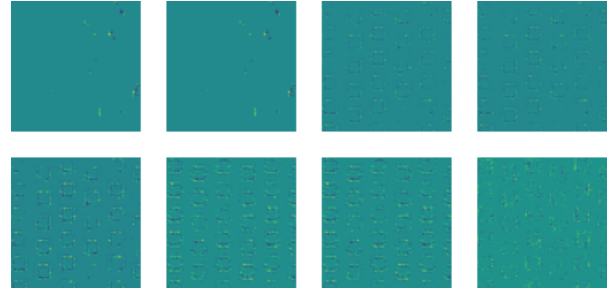


Figure 6: First 8 layers of VGG19 activation layers difference.
HH band - 1.0.

To account for this, a decay penalty is applied across the DWT alpha weights to enforce a descending order, encouraging the model to prioritise watermarking within the LL band, resulting in better robustness against NST.

Algorithm 3: DWT Decay Algorithm

Input:

- DWT alphas (alphas)
-

Output: Decay penalty

```

penalty  0
FOR i  2 TO LENGTH(alphas) DO
    target  alphas[i - 1] *  $\lambda_{DWT}$ 
    penalty  penalty + (alphas[i] - target)
END FOR

```

RETURN penalty

As described in the pseudocode above, the decay loss is calculated using a regularisation penalty, encouraging a smooth decay for each successive DWT layer, dependent on the DWT weight hyperparameter.

Training Process

The watermarking system utilised end-to-end training, wherein all components of the pipeline are performed sequentially. This allowed for direct backpropagation of loss values, both from the image and the adversary detector.

Algorithm 4: Training Process

Input:

- Content Image (I_{con})
 - Style Image (I_{sty})
 - Watermark Image (I_{wm})
 - Epoches (epochs)
-

Output: Best embedding parameters

Initialise *Adversary*()

DWT_alphas = [0.1,0.1,0.1,0.1]

DCT_alpha = 0.1

learning_rate = 0.01

optimiser Adam([*DWT_alphas*, *DCT_alpha*], *learning_rate*)

adv_learning_rate = 0.005

adv_optimiser Adam(*Adversary*(), *adv_learning_rate*)

styled_{ori} perform_nst(*I_{con}*, *I_{sty}*)

FOR RANGE(epochs) **DO**

wm embed_DCT_DWT(*I_{con}*, *I_{wm}*, *DWT_alphas*, *DCT_alpha*)

styled_{wm} perform_nst(*wm*, *I_{sty}*)

adv_prediction *adv_optimiser*(*I_{con}*, *styled_{wm}*)

 Compute loss components with

Equation 3. Equation 2. Equation 1.

 Compute loss with **Equation 4.**

 Compute and add DWT decay with **Algorithm 3.**

 Update *optimiser*

 Update *adv_optimiser*

END FOR

RETURN *DWT_alphas*, *DCT_alpha*

By training the entire system holistically rather than in isolated stages, the encoder could adapt not only to the distortions introduced by the stylisation process but also to adversarial feedback, improving resistance to transformation and detection fidelity.

Experimentation

Watermarking Order



[#figure dct dwt](#): Figure shows effects of performing DCT, or DWT first

Watermark Resolution

To better understand the relationship between watermark size, structure, and robustness against Neural Style Transfer (NST). A series of experiments was conducted to assess the effects of varying resolutions.

An initial hypothesis was formed that watermark imperceptibility and robustness would degrade exponentially with increasing resolution, and by extension, their embedding capacity. Specifically, it was expected that as the watermark's resolution increased, the greater its vulnerability to distortions introduced by Neural Style Transfer; on the other hand, reducing the resolution would help to preserve watermark fidelity by reducing distortions, but limit the amount of information that can be embedded.

This reasoning was based on related works (Luan, 2017; Chang and Chen, 2021; Zhang et al., 2024), which demonstrate that NST introduces distortions across both spatial and frequency domains. In particular, NST heavily modifies high-frequency feature maps. These are the same components typically targeted in traditional watermarking techniques, such as DCT and DWT, to impose the visual characteristics of the style image. As a result, watermarks embedded within these regions are especially prone to degradation.

Given the importance of maintaining watermark visibility after stylistic transformations, understanding this trade-off between embedding capacity and robustness is essential for developing practical, real-world solutions.

To test this, four tests were done using a variable watermark size at 32x32, 16x16, 8x8, and 4x4 to gain insights. Its output can be seen below in **Figure 7**



Figure 7: Watermark variable size: 32x32, 16x16, 8x8, 4,4 (left - right, top - bottom)

To assess metrics, the MSE pixel difference, PSNR, Perceptual Difference, and SSIM were gathered to compare the image. Specifically, the final watermarked styled image was compared with a base plate styled image, the original watermark image, and the watermarked image pre-NST.

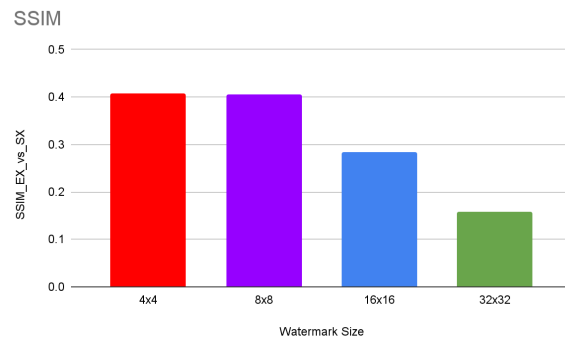
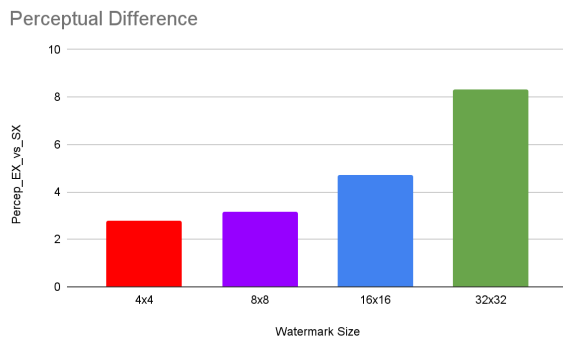
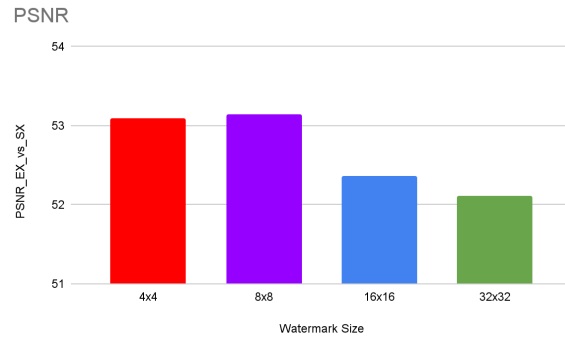
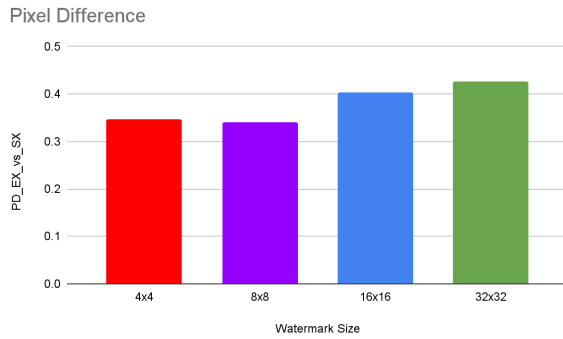


Table 1: Charts showing PSNR, pixel difference, perceptual difference, and structural similarity for varying watermark sizes (4×4, 8×8, 16×16, 32×32). Note: PSNR is rescaled to the range 51-54 to better show the relative difference.

- **Pixel Difference** increases (from ~0.35 to ~0.43) → Extraction becomes less reliable as the watermark size increases.
- **PSNR** decreases (from ~53.08 to ~52.11) → Larger watermarks lose more fidelity.
- **Perceptual Difference** sharply increases (from ~2.77 to ~8.30) → The quality of the extracted watermark after NST degrades significantly with larger sizes.
- **Structural Similarity** drops sharply (from ~0.41 to ~0.15) → The extracted watermark is structurally unrecognizable for large sizes.

As previously mentioned, the empirical watermark used was a 16 x 16 monochromatic watermark. This was chosen as it provides a good trade-off between watermark fidelity and embedding capacity. Additionally, this aligns with prior academic works such as Stylemark (Zhang et al, 2024), providing greater opportunity for comparison and evaluation.

Neural Style Transfer Model

The project initially used the original Neural Style Transfer implementation, proposed by Gatys et al (2016), which introduced the idea of using a CNN to extract and recombine the style of an image, to be used on another. This paper laid the groundwork for all future NST research, making it a logical starting point for research, as it was (naively) thought that since all work branched from this, an implementation based upon this would produce a method more resilient to a wider range of NST models.

This assumption was quickly disproved, however, as Gatys' optimisation approach is shown to be far more computationally expensive compared to more recent works such as AdaIN (Huang and Belongie, 2017), this directly went against the aim of an efficient embedder, as each epoch of the training could take upwards of 30 seconds which, extrapolated over development and experimentation, significantly slowed progress. Alongside this, the fidelity and accuracy of its output far underperformed more recent models. This prompted a shift toward more modern approaches, better suited to an adversarial training framework.

As a result, the project moved to using AdaIN (Huang and Belongie, 2017) as the primary NST implementation. AdaIN's usage of feedforward architecture and statistical style alignment meant the style could be transferred in a single pass, enabling rapid generation and development.

This different implementation also led to drastically different styled results, and different distortions to the watermark as shown in **Figure 8** leading to a watermark that was more visible compared to those stylised by Gatys. This surprising outcome strengthens the need to evaluate robustness against a diverse range of models, as different NST architectures are capable of producing vastly different results.

Gatys



Adain

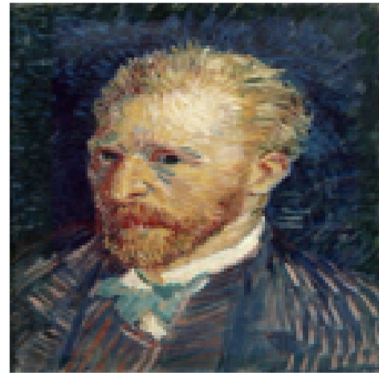
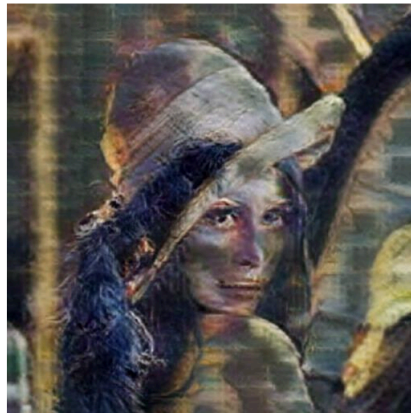
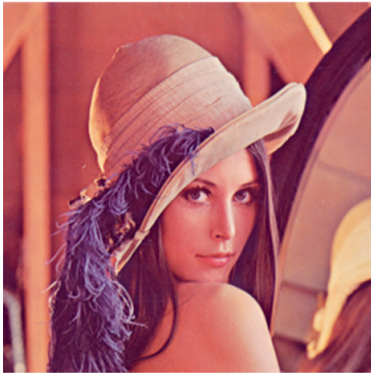


Figure 8: Output difference between Adain and Gatys et al

Results

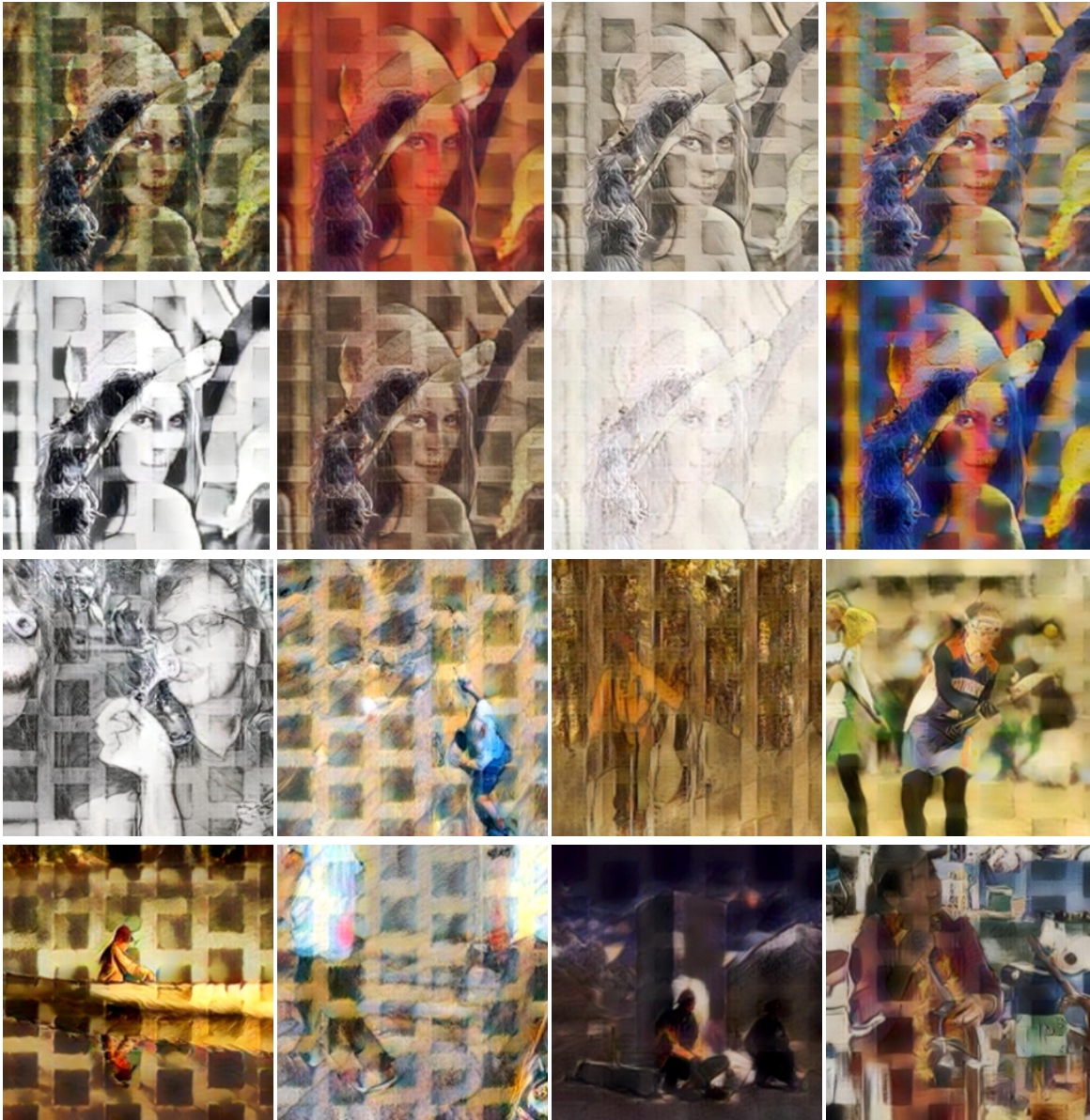


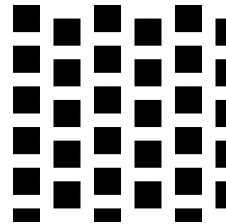
Figure 9: Watermarked image outputs

Analysis

Watermark

One of the central aims was to embed a watermark that could survive the transformations of NST. In this section the robustness and invisibility of the watermark is measured and compared against other implementations including StyleMark.

Original Watermark



Watermarked



Extracted Watermark



Watermark Difference



Figure 10: Degradation of watermark quality, post Neural Style Transfer.

Table 2: Quantitative Differences Between Original Watermark and Extracted Watermark	
Metric	Average Value
Pixel Difference	0.40
PSNR	52.68
Perceptual Difference	5.28
SSIM	0.28

An average pixel difference of 0.4 between the original and extracted watermark confirms a substantial degradation in quality post NST. This reflects the severe impact of stylisation on embedded frequency-domain information, aligning with the observed visual distortions and the inherent instability of watermark preservation under neural style transformations.

This is amplified, due to the image inheriting some colour distortions post NST, a difference of 0.4 indicates an even higher level of distortion.

This is also supported by the perceptual difference with a value of 5.28, obtained by subtracting the difference in feature maps, indicating the watermark image is greatly distorted

Interestingly the Peak Signal to Noise (PSNR) achieves a high value of 52.68dB. This suggests the output input is close in quality to the original stylised image, indicating a less perceptible embedding.

The Structural Similarity Index (SSIM) shows poor structural retention with a mean of 0.28, suggesting the watermark has poor retention post NST. A perfect SSIM score would be 1.0, indicating a perfect structural match, with the result of 0.28 it shows the image has very low structural similarity compared to the reference, and would be perceived as different images by an observer.

Along side this as seen in **Figure 11**, the metrics feature high variability, showing the method used is inconsistent across the images tested, this aligns with visual analysis, as some images do retain the watermark, whilst others it is completely erased. This indicates the method has poor consistency across content images, supporting the idea NST transformations are non-linear and dependent on the image used.



Figure 11: Quantitative range between the original watermark and the extracted watermark

Styled Image

Table 3: Quantitative Differences Between Styled and Watermarked+Styled Images	
Metric	Average Value
Pixel Difference	0.08
PSNR	68.47
Perceptual Difference	0.61
SSIM	0.63



Figure 12: Quantitative range between styled and watermarked+styled images.



Residual

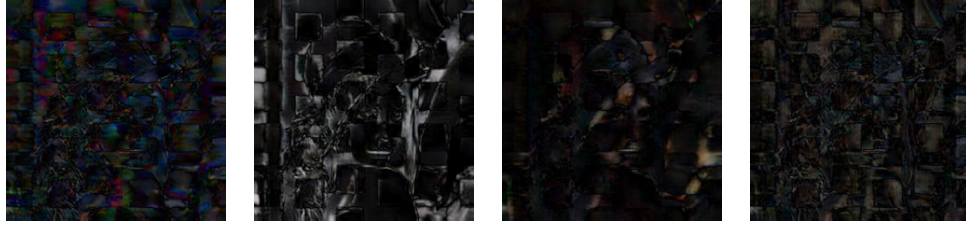


Figure 13: Residual visualisation between the styled image and the watermarked styled image

To further evaluate the impact of watermark embedding on Neural Style Transfer, residual difference images were generated by subtracting the stylised image from the watermarked image; the results can be seen in **Figure 13**

These visualisations reveal significant distortion in both structure and texture between the images, showing strong outlines, altered contours, and visible regions that were not present within the original stylised image, indicating the embedding process is having a more profound impact on the result than expected.

This aligns with prior discussions regarding the non-linear and chaotic nature of NST, specifically, it can be seen that the watermark, which was originally embedded into the isolated frequency domain, has not remained stationary and has been redistributed throughout the image, likely due to NST's usage of deep learning for feature detection.

This “leakage” manifests in residuals as artefacts highlighting regions resembling the watermark image and amplifying broader distortions to the content image’s structure. Such effects are particularly pronounced when the style network has high sensitivity to spatial features, which is often the case with adaptive instance normalization (AdaIN)-based models.

Overall, the residual analysis underscores a key trade-off between invisibility and robustness; while frequency-based embedding can survive aggressive transformations like NST, it often does so at the cost of perceptual subtlety. These findings suggest that future approaches to NST-resilient watermarking must more carefully balance signal strength with visual imperceptibility.

Table 4: Metric comparison between the paper method, Stylemark, and NSP.			
Metric	Method	Stylemark	NSP
PSNR	68.475998	35.632	N/A
SSIM	0.63487	0.952	0.568

To contextualise the effectiveness of the proposed watermarking system, it was compared against two recent studies: Stylemark (Zhang et al. 2024), and NSP (Li et al. 2024). While their implementations vary, PSNR and SSIM gives a shared metric to compare against as shown in **Table 4**

The proposed method produces a significantly higher PSNR of 52.68 dB, compared to the 35.632 DB of Stylemark; on paper, this should mean the method is superior in terms on invisibility compared to others however, these metrics aren't directly comparable as PSNR compares the pixel-level fidelity rather than the robustness and retrievability, which, when utilising different embedding methodologies, can produce non-objective results. Stylemark might take a hit on PSNR because it's embedding a stronger, more recoverable watermark directly within the feature domain, even if it's more noticeable to PSNR, whereas this method produces a more fragile result, utilising the frequency domain that simply gets a better result when evaluated with PSNR.

In contrast, the SSIM score is lower for the proposed method (0.28) than Stylemark (0.952) and higher than NSP (0.568). This suggests that structural consistency is compromised far more when compared to invisible watermarking strategies, and also performs worse as an adversary against NST. One interpretation is that embedding into deep frequency components causes non-obvious structural shifts that get picked up by the feature extraction process of VGG19, significantly distorting the NST result. This is consistent with prior observations of NST's non-linear effects, which may amplify structural distortion of small details.

This divergence reinforces the complexity of evaluating watermark robustness under NST: methods may favour either structural preservation or minimal visual noise, but rarely both, and depending on the overall embedding process or goal of the method, these metrics may not directly correlate between studies.

Future Work

Whilst the method used demonstrates partial robustness to Neural Style Transfer, several areas remain that require further investigation and improvement:

Embedding Method

This method is currently using DWT and DCT for watermark embedding, however as discussed this comes with several issues related to its feature representation, and how the frequency domain gets distorted post NST. Further work is needed to be investigate NSTs effects on the frequency domain, and how it can be counteracted

Improved Extraction

The current extraction method relies on the inverse DWT and DCT to obtain the extracted watermark however, as shown this watermark is unreliable and not always visible (even if visible in the watermarked image), so it could benefit from using either a separate extraction network, or utilising a confidence factor like whats seen in Stylemark.

Multi-Modal Robustness

The current implementation focuses on robustness against a single NST model (AdaIN). Future work could attempt to improve this by incorporating multiple Style Transfer methods, such as CAST (Zhang et al. 2022) and blah blah.

Adversarial Improvements

The current adversary network takes the raw original image and watermarked image as input to detect. This could be improved by instead using the feature map representations as input, as that is what NST works with.

Invisibility Improvements

While robustness is what was prioritised, severe residual distortions were present in the final result which goes against the idea of a robust, invisible watermark. Further refinement needs to be performed to improve the embedding process, one idea could be to extract shared style features to embed the watermark in, like what's performed by Stylemark(Zhang et al. 2024) and HiDDeN(Zhu et al. 2018). This could help with the issues encountered, as it was shown a lower embedding alpha may not be plausible, so utilising some form of masking to limit the effected areas could rectify this.

Robustness Against Other Distortions

While the watermark does perform under NST, no work towards robustness against other distortions were performed notably, other artistic filters, compression, resizing, cropping and so on as laid out in other works.

Conclusion

This study explored the challenges of embedding a robust, imperceptible watermark into an image in the presence of NST, a task that demanded adapting the understanding from traditional watermarking methods, in order to become resilient against style transformations. As a result a hybrid DCT-DWT embedding system inside a machine learning framework was produced.

Initial experimentation with NST revealed the limitations and challenges that a watermark faces against stylisation, notably how embedded signals can become greatly distorted and “bleed” into other areas of the image.

Analysis of the watermark and style image, along with the residual image reveals the challenges of maintaining a frequency domain watermark. Results demonstrated that while the watermark can be embedded to successfully survive NST, the result is unstable and undergoes severe

distortions. Additionally, the findings that this methods PSNR value is lower than Stylemarks, which had a far greater result, brings an interesting question of how to successfully evaluate the watermarks robustness and invisibility, indicating the approach of a frequency domain watermark and loss system used is imperfect.

In summary, this project contributes a novel approach to watermark survival against NST and demonstrates that a traditional watermark such as DCT DWT isn't completely obsolete against it. The work hopes to lay a foundation for future research into the matter, aiming to protect digital content in an era where digital content protection is so vital.

References

- Barni, Mauro, Franco Bartolini, Vito Cappellini, and Alessandro Piva. 1998. "A DCT-Domain System for Robust Image Watermarking." *Signal Processing* 66 (3): 357–72. [https://doi.org/10.1016/s0165-1684\(98\)00015-2](https://doi.org/10.1016/s0165-1684(98)00015-2).
- Chang, Jia-Ren, and Yong-Sheng Chen. 2021. "Exploiting Spatial Relation for Reducing Distortion in Style Transfer." https://openaccess.thecvf.com/content/WACV2021/papers/Chang_Exploiting_Spatial_Relation_for_Reducing_Distortion_in_Style_Transfer_WACV_2021_paper.pdf.
- Cox, I.J., J. Kilian, F.T. Leighton, and T. Shamon. 1997. "Secure Spread Spectrum Watermarking for Multimedia." *IEEE Transactions on Image Processing* 6 (12): 1673–87. <https://doi.org/10.1109/83.650120>.
- fbcotter. 2019. "Pytorch_wavelets (1.3.0)." GitHub. October 3, 2019. https://github.com/fbcotter/pytorch_wavelets.
- Gatys, Leon, Alexander Ecker, and Matthias Bethge. 2016. "A Neural Algorithm of Artistic Style." *Journal of Vision* 16 (12): 326. <https://doi.org/10.1167/16.12.326>.
- Grund, Pihlgren Gustav, Fredrik Sandin, and Marcus Liwicki. 2020. "Improving Image Autoencoder Embeddings with Perceptual Loss." ArXiv.org. January 10, 2020. <https://arxiv.org/abs/2001.03444>.
- Huang, Xun, and Serge Belongie. 2017. "Arbitrary Style Transfer in Real-Time with Adaptive Instance Normalization." *ArXiv:1703.06868 [Cs]*, July. <https://arxiv.org/abs/1703.06868>.
- Icaro. 2019. "Best Artworks of All Time." Kaggle.com. 2019. <https://www.kaggle.com/datasets/ikarus777/best-artworks-of-all-time/data>.

- Jia, Zhaoyang, Han Fang, and Weiming Zhang. 2021. "MBRS: Enhancing Robustness of DNN-Based Watermarking by Mini-Batch of Real and Simulated JPEG Compression." *Proceedings of the 29th ACM International Conference on Multimedia*, October, 41–49. <https://doi.org/10.1145/3474085.3475324>.
- Jing, Yongcheng, Yezhou Yang, Zunlei Feng, Jingwen Ye, Yizhou Yu, and Mingli Song. 2020. "Neural Style Transfer: A Review." *IEEE Transactions on Visualization and Computer Graphics* 26: 3365–85. <https://doi.org/10.1109/TVCG.2019.2921336>.
- Kavitha, R S, U Eranna, and M N Giriprasad. 2020. "DCT-DWT Based Digital Watermarking and Extraction Using Neural Networks," 1–5. <https://doi.org/10.1109/AISP48273.2020.9073104>.
- Kulkarni, Gururaj, and Suresh Kuri. 2017. "Robust Digital Image Watermarking Using DWT, DCT and Probabilistic Neural Network." *2017 International Conference on Electrical, Electronics, Communication, Computer, and Optimization Techniques (ICEECOT)*, December. <https://doi.org/10.1109/iceecot.2017.8284571>.
- Lee, Jae-Eun, Young-Ho Seo, and Dong-Wook Kim. 2020. "Convolutional Neural Network-Based Digital Image Watermarking Adaptive to the Resolution of Image and Watermark." *Applied Sciences* 10. <https://doi.org/10.3390/app10196854>.
- Li, Yaxin, Jie Ren, Han Xu, and Hui Liu. 2024. "Neural Style Protection: Counteracting Unauthorized Neural Style Transfer," 3954–63. <https://doi.org/10.1109/WACV57701.2024.00392>.
- Li, Yue, Hongxia Wang, and Mauro Barni. 2021. "A Survey of Deep Neural Network Watermarking Techniques." ArXiv.org. 2021. <https://arxiv.org/abs/2103.09274>.
- Luan, Fujun, Sylvain Paris, Eli Shechtman, and Kavita Bala. 2017. "Deep Photo Style Transfer." ArXiv.org. March 22, 2017. <https://arxiv.org/abs/1703.07511>.
- Mallat, S.G. 1989. "A Theory for Multiresolution Signal Decomposition: The Wavelet Representation." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 11: 674–93. <https://doi.org/10.1109/34.192463>.
- Marshall, Dave. 2001. "The Discrete Cosine Transform (DCT)." Users.cs.cf.ac.uk. Cardiff University. 2001. <https://users.cs.cf.ac.uk/dave/Multimedia/node231.html>.
- Navas, K. A., Mathews Cheriyan Ajay, M. Lekshmi, Tampy S. Archana, and M. Sasikumar. 2008. "DWT-DCT-SVD Based Watermarking." *IEEE Xplore*. January 1, 2008. <https://doi.org/10.1109/COMSWA.2008.4554423>.

- Simonyan, Karen, and Andrew Zisserman. 2015. "Very Deep Convolutional Networks for Large-Scale Image Recognition." ArXiv.org. April 10, 2015. <https://arxiv.org/abs/1409.1556>.
- Specht, Donald F. 1990. "Probabilistic Neural Networks." *Neural Networks* 3 (1): 109–18. [https://doi.org/10.1016/0893-6080\(90\)90049-q](https://doi.org/10.1016/0893-6080(90)90049-q).
- Swanson, Mitchell D, Bin Zhu, Ahmed H Tewfik, and Laurence Boney. 1998. "Robust Audio Watermarking Using Perceptual Masking." *Signal Processing* 66 (3): 337–55. [https://doi.org/10.1016/s0165-1684\(98\)00014-0](https://doi.org/10.1016/s0165-1684(98)00014-0).
- Tavakoli, Alireza, Zahra Honjani, and Hedieh Sajedi. 2022. "Convolutional Neural Network-Based Image Watermarking Using Discrete Wavelet Transform." ArXiv.org. 2022. <https://arxiv.org/abs/2210.06179>.
- Tirkel, Andrew, G.A Rankin, Ron G. van Schyndel, W. J. Ho, N. Mee, and C. F. Osborne. 1994. "Electronic Watermark." In *Digital Image Computing, Technology and Applications*.
- Ulyanov, Dmitry, Vadim Lebedev, Andrea Vedaldi, and Victor Lempitsky. 2016. "Texture Networks: Feed-Forward Synthesis of Textures and Stylized Images." ArXiv.org. March 10, 2016. <https://doi.org/10.48550/arXiv.1603.03417>.
- Wang, Quan, Sheng Li, Xinpeng Zhang, and Guorui Feng. 2023. "Rethinking Neural Style Transfer: Generating Personalized and Watermarked Stylized Images." *Proceedings of the 31st ACM International Conference on Multimedia*, October, 6928–37. <https://doi.org/10.1145/3581783.3612202>.
- Wang, Zhou, A.C. Bovik, H.R. Sheikh, and E.P. Simoncelli. 2004. "Image Quality Assessment: From Error Visibility to Structural Similarity." *IEEE Transactions on Image Processing* 13 (4): 600–612. <https://doi.org/10.1109/tip.2003.819861>.
- Xia, Xiang-Gen, Charles G Boncelet, and Gonzalo R Arce. 1998. "Wavelet Transform Based Watermark for Digital Images." *Opt. Express* 3: 497–511. <https://doi.org/10.1364/OE.3.000497>.
- Xue, Yuliang, Nan Zhong, Zhenxing Qian, and Xinpeng Zhang. 2022. "PSTNet: Protectable Style Transfer Network Based on Steganography," 54–59. <https://doi.org/10.1109/CoST57098.2022.00021>.
- zh217. 2018. "Torch-Dct (0.1.6)." GitHub. 2018. <https://github.com/zh217/torch-dct>.
- Zhang, Yunming, Dengpan Ye, Sipeng Shen, and Jun Wang. 2024. "StyleMark: A Robust Watermarking Method for Art Style Images against Black-Box Arbitrary Style Transfer." ArXiv.org. 2024. <https://arxiv.org/abs/2412.07129>.

Zhang, Yuxin, Fan Tang, Weiming Dong, Haibin Huang, Chongyang Ma, Tong-Yee Lee, and Changsheng Xu. 2022. "Domain Enhanced Arbitrary Image Style Transfer via Contrastive Learning." *Special Interest Group on Computer Graphics and Interactive Techniques Conference Proceedings*, August, 1–8.

<https://doi.org/10.1145/3528233.3530736>.

Zhu, Jiren, Russell Kaplan, Justin Johnson, and Li Fei-Fei. 2018. "HiDDeN: Hiding Data with Deep Networks." ArXiv.org. 2018. <https://arxiv.org/abs/1807.09937>.